Spin representation via bosons

SHO $\quad [\hat{x}, \hat{p}] = i\hbar$

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}\,(a^+ + a)$$

$$\hat{p} = i\sqrt{\frac{\hbar m\omega}{2}}\,(a^+ - a)$$

$$[\hat{x}, \hat{p}] = \frac{i\hbar}{2}\,[a^+ + a,\ a^+ - a] = \frac{i\hbar}{2}\Big([\cancel{a^+, a^+}] - [a^+, a] + [a, a^+] - [\cancel{a, a}]\Big) = i\hbar \quad \text{when } [a, a^+] = 1$$

$$\big(\Rightarrow [a^+, a] = -1\big).$$

it follows that
$$\begin{cases} a^+ |n\rangle = \sqrt{n+1}\ |n+1\rangle \\ a\ |n\rangle = \sqrt{n}\ |n-1\rangle \end{cases}$$

then $a^+ a\ |n\rangle = a^+\big(\sqrt{n}\ |n-1\rangle\big) = \sqrt{n}\ \sqrt{n-1+1}\ |n-1+1\rangle = n\,|n\rangle$

$\Rightarrow \quad a^+ a = \hat{n}$ = number operator

Let us now introduce another pair of boson, b and b^+:

$[b, b^+] = 1$ but $[b, a] = [b, a^+] = 0$

let
$$(1)\quad \begin{cases} S^z = \frac{\hbar}{2}\,(a^+ a - b^+ b) \\ S^+ = \hbar a^+ b \qquad S^x = \frac{1}{2}(S^+ + S^-) = \frac{\hbar}{2}(a^+ b + b^+ a) \\ S^- = \hbar b^+ a \qquad S^y = \frac{1}{2i}(S^+ - S^-) = \frac{\hbar}{2}(a^+ b - b^+ a) \end{cases}$$

Check identity ① $[S^+, S^-] = [S^x + iS^y,\ S^x - iS^y] = -i\,[S^x, S^y] + i\,[S^y, S^x] = -i\ i\hbar S^z + i(-i\hbar S^z) = 2\hbar S^z$

$$[a^+b, b^+a] = a^+ \underbrace{b b^+}_{1+b^+b} a - b^+ \underbrace{a a^+}_{1+a^+a} b =$$

$$= a^+a + a^+a\, b^+b - (b^+b + b^+b\, a^+a) = a^+a - b^+b$$

$$\Rightarrow \hbar^2 [a^+b, b^+a] = 2\hbar \cdot \frac{\hbar}{2}(a^+a - b^+b) = 2\hbar\, S^z.$$

② $[S^z, S^x] = i\hbar\, \underset{+1}{\underset{\parallel}{\varepsilon^{zxy}}} S^y$

$$\frac{\hbar^2}{4}[a^+a - b^+b,\ a^+b + b^+a] = \frac{\hbar^2}{4}\Big([a^+a, a^+b + b^+a]$$

$$- [b^+b, a^+b + b^+a]\Big) = \frac{\hbar^2}{4}\big(a^+b - b^+a - (b^+a - a^+b)\big)$$

because $[a^+a, a^+b] = a^+ \underbrace{a a^+}_{1+a^+a} b - a^+b\, a^+a = a^+(1+a^+a)b - a^+a^+ab = a^+b$

$$[a^+a, b^+a] = a^+a\, b^+a - b^+a\, a^+a = b^+a^+aa - b^+(1+a^+a)a =$$

$$= -b^+a$$

$$\Rightarrow \frac{\hbar^2}{4}[a^+a - b^+b, a^+b + b^+a] = \frac{\hbar^2}{2}(a^+b - b^+a) =$$

$$= i\hbar \cdot \frac{\hbar}{2i}(a^+b - b^+a) = i\hbar\, S^y \quad !$$

③ Obviously $[S^z, S^y] = -i\hbar S^x$.

We also know that in the basis of angular momentum eigenstates $\hat{S}^2 = (S^x)^2 + (S^y)^2 + (S^z)^2 = \hbar^2 s(s+1)$

$$(S^x)^2 + (S^y)^2 = \left(\frac{S^+ + S^-}{2}\right)^2 + \left(\frac{S^+ - S^-}{2i}\right)^2 =$$

$$= \frac{1}{4}\big(S^+S^+ + S^-S^- + S^+S^- + S^-S^+ - S^+S^+ - S^-S^-$$

$$+ S^+S^- + S^-S^+\big) = \frac{1}{2}(S^+S^- + S^-S^+)$$

$$\Rightarrow \hat{S}^2 = \frac{\hbar^2}{4}\left(a^+a - b^+b\right)\left(a^+a - b^+b\right) + \frac{\hbar^2}{2}\left(a^+b\, b^+a + \right.$$

$$\left. + b^+a\, a^+b\right) = \frac{\hbar^2}{4}\Big\{ a^+a\, a^+a + b^+b\, b^+b$$

$$- a^+a\, b^+b - b^+b\, a^+a + 2\, a^+\underbrace{b\, b^+}_{1+b^+b}a + 2b^+b\,\underbrace{a\, a^+}_{1+a^+a}\Big\}$$

$$= \frac{\hbar^2}{4}\Big\{ \overset{a^+aa^+a}{\cancel{a^+(1+a^+a)a}} + \overset{b^+bb^+b}{\cancel{b^+(1+b^+b)b}} + a^+a\, b^+b + b^+b\, a^+a$$

$$+ 2\left(a^+a + b^+b\right)\Big\} =$$

$$= \frac{\hbar^2}{4}\left(a^+a + b^+b\right)\left(a^+a + b^+b + 2\right) = \hbar^2 S(S+1)$$

Thus $\hat{S}^2 = \hbar^2 S(S+1)$ only if we <u>require</u>

$$\boxed{a^+a + b^+b = 2S} \qquad (2)$$

* Equations (1) and (2) form Schwinger boson representation of spin operator $\hat{S}$.

<u>Holstein – Primakoff representation</u> (H-P)

Let us "resolve" constraint (2) for b^+b:

$$b^+b = 2S - a^+a = \sqrt{2S - a^+a}\cdot\sqrt{2S - a^+a}$$

and require

$$b = b^+ = \sqrt{2S - a^+a} \quad \text{so that} \quad \boxed{a^+a \le 2S}$$

then $S^z = \frac{\hbar}{2}\left(a^+a - (2S - a^+a)\right) = \frac{\hbar}{2}\left(2a^+a - 2S\right) =$

$$= a^+a - S$$

$$(3)\quad \begin{cases} S^z = a^+a - S \\ S^+ = \hbar\, a^+\sqrt{2S - a^+a} \\ S^- = \hbar\sqrt{2S - a^+a}\, a \end{cases}$$

Could also do: $a^+a = 2S - b^+b = \sqrt{2S - b^+b}\cdot\sqrt{2S - b^+b}$

[better choice] with $\underline{b^+b \le 2S}$

$$\Rightarrow\quad (3')\quad \begin{cases} S^z = \frac{\hbar}{2}\left(2S - b^+b - b^+b\right) = \hbar\left(S - b^+b\right). \\ S^+ = \hbar\sqrt{2S - b^+b}\, b \\ S^- = \hbar\, b^+\sqrt{2S - b^+b} \end{cases}$$

$$S^+ = \sqrt{2S}\left(1 - \frac{b^+b}{2S}\right)^{1/2} b = \sqrt{2S}\left(1 - \frac{b^+b}{4S} + \dots\right) b.$$

$$S^- = \sqrt{2S}\, b^+\left(1 - \frac{b^+b}{4S}\right).$$

Can be checked that (3) and (3') satisfy spin commutation relations, for as long as $\begin{cases} a^+a \le 2S \\ b^+b \le 2S. \end{cases}$

H-P transformation is useful for <u>semi-classical</u> expansions about the ordered states, while Schwinger bosons are used for studying (quantum and thermal) disordered spin states.

Indeed, let us treat $\frac{b^+b}{2S}$ as a small parameter

$$\boxed{\frac{b^+b}{2S} \ll 1}$$

Then $S^z = \hbar\left(S - b^+b\right)$

$$S^+ = \hbar\sqrt{2S}\sqrt{1 - \frac{b^+b}{2S}}\; b \approx \hbar\sqrt{2S}\left(1 - \frac{b^+b}{4S} + \dots\right) b$$

$$= \hbar\sqrt{2S}\left(b - \frac{b^+b\, b}{4S} + \dots\right)$$

$$S^- = \hbar\, b^+ \sqrt{2S}\sqrt{1-\frac{b^+b}{2S}} \approx \hbar\sqrt{2S}\, b^+\left(1-\frac{b^+b}{4S}+\dots\right)$$

$$= \hbar\sqrt{2S}\left(b^+ - \frac{b^+b^+b}{4S}+\dots\right)$$

Notice that now we can <u>approximate</u>

$$S^+ = \hbar\sqrt{2S}\, b$$

$$S^- = \hbar\sqrt{2S}\, b^+$$

$$\Rightarrow [S^+, S^-] = \hbar^2(2S)\underbrace{[b, b^+]}_{1} = 2\hbar\cdot\hbar S$$

$$= 2\hbar\cdot \cancel{\hbar(S-b^+b)}\ \hbar S\left(1-\frac{b^+b}{S}\right) \approx 2\hbar\cdot\hbar S$$

i.e. the commutation relation is <u>okay</u> for as long as number of bosons <u>is small</u> : $\frac{b^+b}{S} \ll 1$.

Will see later that this has the meaning of "small" deviations from the classical ground state.

Many spins: Heisenberg ferromagnet

Spins from different sites commute $[S^a(r), S^b(r')] = 0$ for $r \neq r'$.

$$H = -\frac{1}{2} J \sum_{n,g} \vec{S}_n \cdot \vec{S}_{n+g}$$

For any pair of spins: $\vec{S}_n \cdot \vec{S}_{n+g} = S_n^z \cdot S_{n+g}^z + \frac{1}{2}(S_n^+ S_{n+g}^- + S_n^- S_{n+g}^+)$

$$= (S - b_n^+ b_n)(S - b_{n+g}^+ b_{n+g}) + \frac{1}{2}(2S)(b_n b_{n+g}^+ + b_n^+ b_{n+g}) =$$

$$= S^2 - S(b_n^+ b_n + b_{n+g}^+ b_{n+g} - b_n^+ b_{n+g} - b_{n+g}^+ b_n)$$
$$+ b_n^+ b_n \cdot b_{n+g}^+ b_{n+g}$$

$$H = -\frac{1}{2} J \Big\{ \underbrace{S^2 \cdot zN}_{\substack{\text{classical energy} \\ \text{(every spin has } z \text{ neighbors)}}} - S \sum_{n,g} (b_n^+ b_n + b_{n+g}^+ b_{n+g} - b_n^+ b_{n+g} - b_{n+g}^+ b_n)$$
$$+ \sum_{ng} \underbrace{b_n^+ b_n \cdot b_{n+g}^+ b_{n+g}}_{\text{interaction between spin waves}} \Big\}$$

Observe that last term does not have factor (S) in front of the sum $\Rightarrow$ it gives $\frac{1}{S}$ contribution in comparison with the second sum

$\Rightarrow$ small for large spins, $\underline{S \gg 1}$

$$H = H^{(0)} + H^{(2)} + H^{(4)}$$

where
$$\left.\begin{array}{ll} H^{(0)} & \sim S^2 \\ H^{(2)} & \sim S \\ H^{(4)} & \sim O(1) \end{array}\right\} \begin{array}{l} \rightarrow \text{linear spin wave approximation} \\ \rightarrow \text{neglect } O(1) \text{ contribution!} \end{array}$$

$$H \approx H^{(0)} + H^{(2)}.$$

$$H^{(2)} = \frac{1}{2} JS \sum_{ng} \left(2 b_n^+ b_n - b_n^+ b_{n+g} - b_{n+g}^+ b_n \right)$$

Fourier transform to momentum space:

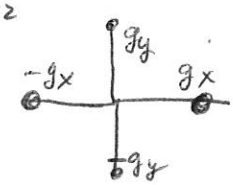
$$b_n = \frac{1}{\sqrt{N}} \sum_k e^{i\bar{k}\bar{R}_n} b_k$$

$$b_n^+ = \frac{1}{\sqrt{N}} \sum_k e^{-i\bar{k}\bar{R}_n} b_k^+$$

$$H^{(2)} = \frac{1}{2} JS \frac{1}{N} \sum_{k_1 k_2} \sum_{ng} \Big(2 b_{k_1}^+ b_{k_2} e^{-i\bar{R}_n(\bar{k}_1 - \bar{k}_2)}$$

$$- e^{-i\bar{R}_n(\bar{k}_1-\bar{k}_2)} e^{+i\bar{g}\bar{k}_1} b_{k_1}^+ b_{k_2} - e^{-i\bar{R}_n(\bar{k}_1-\bar{k}_2)} e^{-i\bar{g}\bar{k}_1} b_{k_1}^+ b_{k_2} \Big)$$

$$= \frac{1}{2} JS \frac{1}{N} \sum_{k_1 k_2} \underbrace{\sum_{R_n} e^{-i\bar{R}_n(\bar{k}_1-\bar{k}_2)}}_{N\Delta_{\bar{k}_1,\bar{k}_2} \Rightarrow \bar{k}_1 = \bar{k}_2} \sum_g b_{k_1}^+ b_{k_2} (2 - 2\cos \bar{g}\cdot\bar{k}_1)$$



$$= JS \sum_k \varepsilon_k b_k^+ b_k$$

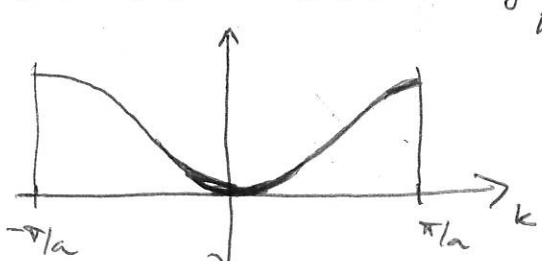
where $\varepsilon_k = \sum_g (1 - \cos \bar{k}\cdot\bar{g}) = 2(1-\cos k_x a) + 2(1-\cos k_y a) + 2(1-\cos k_z a) =$

$$= 6 - 2(\cos k_x a + \cos k_y a + \cos k_z a)$$

Spin wave dispersion : $JS\,\varepsilon(k) \to SJa^2 \vec{k}^2$ for small momenta

<u>min</u> $\varepsilon(0) = 0$

<u>max</u> $\varepsilon(\vec{\pi}) = 12$



$\cos ka \approx 1 - \frac{1}{2}k^2a^2$ for $k \to 0$

$$\Rightarrow \varepsilon(k) \approx 6 - 2\left(3 - \frac{a^2}{2}(k_x^2 + k_y^2 + k_z^2)\right) = a^2 \vec{k}^2 + \ldots$$

quadratic dispersion near the origin!

Thermodynamic limit

$$\frac{1}{N}\sum_k \cdots = v_0 \int_{BZ} \frac{d\vec{k}}{(2\pi)^D} \cdots$$

volume of the lattice unit cell

$$N \cdot v_0 = V = \text{volume}$$

$$H^{(2)} = \sum_k JS\varepsilon(k)\, b_k^+ b_k \quad = \text{ideal gas of bosons.}$$

Number of bosons with momentum $\vec{k}$ at temperature T:

$$n_k = \langle b_k^+ b_k \rangle = \frac{1}{e^{\frac{JS\varepsilon(k)}{k_B T}} - 1}$$ Bose-Einstein distribution

average over thermal ensemble

$$\langle H^{(2)} \rangle = E^{(2)} = \sum_k JS\,\varepsilon(k) \langle b_k^+ b_k \rangle$$

$$= \underbrace{N \cdot v_0}_{\text{Volume}} \int \frac{d\vec{k}}{(2\pi)^3} \frac{JS\varepsilon(k)}{\exp\left(\frac{JS\varepsilon(k)}{k_B T}\right) - 1}$$

But we expect bosons with small $\vec{k}$ (lowest energy) to contribute most $\Rightarrow$ $SJ\varepsilon(k) \approx D\vec{k}^2$ ($D = SJa^2$ = spin stiffness)

$$E^{(2)} = V \int \frac{d\vec{k}}{(2\pi)^3} \frac{D\vec{k}^2}{e^{\frac{D\vec{k}^2}{k_B T}} - 1} =$$

$$= V \cdot \frac{4\pi}{(2\pi)^3} \int_0^\infty dk \cdot k^2 \frac{Dk^2}{e^{\frac{Dk^2}{k_B T}} - 1} =$$

$$= \left(k = \sqrt{\frac{k_B T}{D}}\, x\right) = \frac{V}{2\pi^2} \left(\frac{k_B T}{D}\right)^{\frac{1}{2} \cdot 5} \int_0^\infty \frac{dx\, x^4}{e^x - 1}$$

$$\Rightarrow E^{(2)} \sim T^{5/2}$$

$$C_V = \frac{1}{V}\left(\frac{\partial E}{\partial T}\right)_V \sim T^{3/2}$$

Total number of bosons

$$\Delta N = \sum_k n_k = V \int \frac{d\vec{k}}{(2\pi)^3} \frac{1}{e^{\frac{Dk^2}{k_B T}} - 1} =$$

$$\cong \frac{V}{2\pi^2} \int_0^\infty dk \cdot k^2 \frac{1}{\exp\left(\frac{Dk^2}{k_B T}\right) - 1}$$

$$= \frac{V}{2\pi^2} \left(\frac{k_B T}{D}\right)^{3/2} \int_0^\infty \frac{dx\, x^2}{e^x - 1}$$

Every boson reduces S_{tot} by 1

$\Rightarrow$ on-site ~~magnetic~~ spin $\langle S^z(r) \rangle = S - \langle b_r^+ b_r \rangle$

$$= S - \frac{1}{N} \sum_k n_k = S - (\#) \cdot T^{3/2}$$

Bloch's law.

$\uparrow \downarrow \uparrow$ $\quad J>0$ interaction between nearest spins

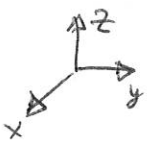
$\downarrow \uparrow \downarrow$ $\quad$ on a cubic lattice.

A sublattice: (up-spin)

$$\begin{cases} S_n^z = S - a_n^+ a_n \\ S_n^+ \cong \sqrt{2S}\, a_n \\ S_n^- \cong \sqrt{2S}\, a_n^+ \end{cases}$$

B sublattice: (down spin)

can obtain by rotation about $\hat{x}$ axis



$$\left.\begin{array}{l} S^x \to S^x \\ S^y \to -S^y \\ S^z \to -S^z \end{array}\right\} \quad \begin{array}{l} S^+ = S^x + iS^y \to S^x - iS^y = S^- \\ S^- = S^x - iS^y \to S^x + iS^y = S^+ \end{array} \Rightarrow \begin{cases} S_m^z = -S + b_m^+ b_m \\ S_m^+ = \sqrt{2S}\, b_m^+ \\ S_m^- = \sqrt{2S}\, b_m \end{cases}$$

$$H = \frac{1}{2} J \sum_{nm} \vec{S}_n \cdot \vec{S}_m = J \sum_{\substack{n \in A \\ (n+g = m \in B)}} \vec{S}_n \cdot \vec{S}_{n+g} =$$

$$= J \sum_{n,g} S_n^z S_{n+g}^z + \frac{1}{2}\left(S_n^+ S_{n+g}^- + S_n^- S_{n+g}^+\right) =$$

$$= J \sum_{n,g} \left(S - a_n^+ a_n\right)\left(-S + b_{n+g}^+ b_{n+g}\right) + \frac{1}{2}(2S)\left(a_n b_{n+g} + a_n^+ b_{n+g}^+\right) = J \sum_{ng} \Big\{ -S^2 + S\left(a_n^+ a_n + b_{n+g}^+ b_{n+g} + a_n b_{n+g} + a_n^+ b_{n+g}^+\right) - \underbrace{a_n^+ a_n b_{n+g}^+ b_{n+g}}_{\text{neglect for now}} \Big\}$$

$$\sum_{ng} (-S^2) = (-S^2)\left(\frac{N}{2}\right)(z)$$

$\frac{N}{2}$: # sites in A sublattice

z: # neighbors $\begin{cases} 6 & \text{for } d=3 \quad \text{cubic} \\ 4 & \text{for } d=2 \quad \text{square} \\ 2 & \text{for } d=1 \quad \text{chain} \end{cases} = \underline{2d}$

Fourier transform: $a_n = \sqrt{\frac{2}{N}} \sum_k e^{-i\vec{k}\vec{R}_n} a_k$

$$b_m = \sqrt{\frac{2}{N}} \sum_k e^{-i\vec{k}\vec{R}_m} b_k$$

where $\sum_k$ is over the magnetic Brillouin zone = Brillouin zone of a lattice formed by A-sublattice only. (consists of $N/2$ sites)

$$H^{(2)} = JS \sum_{n g} \frac{2}{N} \sum_{k_1 k_2} \Big\{ a^+_{k_1} a_{k_2} e^{i\vec{k}_1\vec{R}_n - i\vec{k}_2\vec{R}_n} + b^+_{k_1} b_{k_2} e^{i\vec{k}_1(\vec{R}+\vec{g}) - i\vec{k}_2(\vec{R}+\vec{g})}$$

$$+ a^+_{k_1} b^+_{k_2} e^{i\vec{k}_1\vec{R}_n} e^{i\vec{k}_2(\vec{R}_n+\vec{g})} + a_{k_1} b_{k_2} e^{-i\vec{R}_n(\vec{k}_1+\vec{k}_2)} e^{-i\vec{k}_2\vec{g}} \Big\}$$

$$= JS \sum_g \sum_k \Big\{ a^+_k a_k + b^+_k b_k + a^+_k b^+_{-k} e^{-i\vec{k}\vec{g}} + a_k b_{-k} e^{i\vec{k}\vec{g}} \Big\}$$

$$\sum_{R_n} e^{i\vec{R}_n(\vec{k}_1 \pm \vec{k}_2)} = \frac{N}{2} \begin{cases} \Delta_{k_1+k_2,0} \\ \Delta_{k_1-k_2,0} \end{cases}.$$

anomalous terms (involve two creation/annihilation operators at once.)

$$H^{(2)} = JS \sum_k A_k (a^+_k a_k + b^+_k b_k) + B_k (a^+_k b^+_{-k} + a_{-k} b_k)$$

Here $A_k = z = 2d$; $B_k = \sum_g e^{-i\vec{k}\vec{g}} = 2(\cos k_x a + \cos k_y a + \cos k_z a) = B_{-k}$

<u>Bogoliubov transformation</u>

$$a_k = ch\theta_k \, \alpha_k - sh\theta_k \, \beta^+_{-k} \qquad \text{where } [\alpha_k, \alpha^+_{k'}] = \delta_{kk'}$$

$$b_k = -sh\theta_k \, \alpha^+_k + ch\theta_k \, \beta_{-k} \qquad [\beta_k, \beta^+_{k'}] = \delta_{k,k'}$$

$$[\alpha_k, \beta_{k'}] = 0.$$

Check $[a_k, a^+_{k'}] = [ch\theta_k \, \alpha_k - sh\theta_k \, \beta^+_{-k}, ch\theta_{k'} \, \alpha^+_{k'} - sh\theta_{k'} \, \beta^-_{-k'}]$

$$= ch^2\theta_k \underbrace{[\alpha_k, \alpha^+_{k'}]}_{\delta_{k,k'}} + sh^2\theta_k \underbrace{[\beta^+_{-k}, \beta_{-k'}]}_{-\delta_{k,k'}} = \underbrace{(ch^2\theta_k - sh^2\theta_k)}_{1} \delta_{k,k'}$$

$$\frac{H^{(2)}}{JS} = \sum_k A_k \Big[(ch\theta_k \alpha_k^+ - sh\theta_k \beta_{-k})(ch\theta_k \alpha_k - sh\theta \beta_{-k}^+)$$

$$+ (-sh\theta_k \alpha_k + ch\theta_k \beta_{-k}^+)(-sh\theta_k \alpha_k^+ + ch\theta_k \beta_{-k}) \Big]$$

$$+ B_k \Big[(ch\theta_k \alpha_k^+ - sh\theta_k \beta_{-k})(-sh\theta_k \alpha_k + ch\theta_k \beta_{-k}^+) +$$

$$+ (ch\theta_k \alpha_k - sh\theta_k \beta_{-k}^+)(-sh\theta_k \alpha_k^+ + ch\theta_k \beta_{-k}) \Big] =$$

$$= \sum_k \Big[A_k (ch^2\theta_k \overbrace{+ sh^2\theta_k}^{\alpha_k^+\alpha_k}) \overbrace{}^{\alpha_k\alpha_k^+} - B_k ch\theta_k sh\theta_k (\alpha_k^+\alpha_k + \alpha_k\alpha_k^+) \Big]$$

$$+ \Big[A_k (sh^2\theta_k \overbrace{+ ch^2\theta_k}^{\beta_{-k}\beta_{-k}^+} \overbrace{}^{\beta_{-k}^+\beta_{-k}}) - B_k sh\theta_k ch\theta_k (\beta_{-k}\beta_{-k}^+ + \beta_{-k}^+\beta_{-k}) \Big] +$$

$$- \alpha_k^+ \beta_{-k}^+ \Big[2A_k ch\theta_k sh\theta_k - B_k (ch^2\theta_k + sh^2\theta_k) \Big]$$

$$- \alpha_k \beta_{-k} \Big[2A_k ch\theta_k sh\theta_k - B_k (sh^2\theta_k + ch^2\theta_k) \Big] .$$

We <u>require</u> the anomalous terms $\alpha_k^+ \beta_{-k}^+$, $\alpha_k \beta_{-k}$ <u>to be absent:</u>

$$\Rightarrow \quad A_k sh 2\theta_k - B_k ch 2\theta_k = 0$$

$$\Rightarrow \quad \frac{sh 2\theta_k}{ch 2\theta_k} = \frac{B_k}{A_k} .$$

$$\Rightarrow H^{(2)}/JS = \sum_k \Big(A_k ch 2\theta_k - \tfrac{1}{2} B_k sh 2\theta_k \cdot 2 \Big) \alpha_k^+ \alpha_k - \tfrac{1}{2} B_k sh 2\theta_k$$

$$+ \Big(A_k ch 2\theta_k - B_k sh 2\theta_k \Big) \beta_{-k}^+ \beta_{-k} - \tfrac{1}{2} B_k sh 2\theta_k$$

$$- 2A_k sh^2\theta_k$$

$\alpha_k \alpha_k^+ =$
$= 1 + \alpha_k^+ \alpha_k$

let $\omega_k = A_k ch 2\theta_k - B_k sh 2\theta_k$

$$ch^2 2\theta - sh^2 2\theta = 1 \Rightarrow \frac{sh\, 2\theta_k}{\sqrt{1+sh^2 2\theta_k}} = \frac{B_k}{A_k} \quad \left(= \text{even function of } \vec{k}\right)$$

$$\frac{sh^2 2\theta_k}{1+sh^2 2\theta_k} = \frac{B_k^2}{A_k^2} \Rightarrow \left(1 - \frac{B_k^2}{A_k^2}\right) sh^2 2\theta_k = \frac{B_k^2}{A_k^2}$$

$$\Rightarrow sh^2 2\theta_k = \frac{B_k^2}{A_k^2 - B_k^2}$$

$$\Rightarrow ch^2 2\theta_k = 1 + \frac{B_k^2}{A_k^2 - B_k^2} = \frac{A_k^2}{A_k^2 - B_k^2}$$

$$sh\, 2\theta_k = sgn(B_k) \frac{|B_k|}{\sqrt{A_k^2 - B_k^2}}, \quad B_k\, sh\, 2\theta_k = \frac{B_k^2}{\sqrt{A_k^2 - B_k^2}}$$

$$ch\, 2\theta_k = \frac{A_k}{\sqrt{A_k^2 - B_k^2}}$$

$$\omega_k = \frac{A_k^2 - B_k^2}{\sqrt{A_k^2 - B_k^2}} = \sqrt{A_k^2 - B_k^2}$$

Constant term: $2A_k\, sh^2\theta_k - B_k\, sh\, 2\theta_k = A_k\, ch\, 2\theta_k - B_k\, sh\, 2\theta_k - A_k$

$$= \omega_k - A_k$$

$$2\, sh^2\theta_k = (ch\, 2\theta_k - 1)$$

$$\Rightarrow H^{(2)} = JS \sum_k \left\{ \omega_k \left(\alpha_k^+ \alpha_k + \beta_k^+ \beta_k\right) + \omega_k - A_k \right\}$$

$$= JS \sum_k \left\{ \omega_k \left(\alpha_k^+ \alpha_k + \tfrac{1}{2}\right) + \omega_k \left(\beta_k^+ \beta_k + \tfrac{1}{2}\right) \right\}.$$

$$\underbrace{- JS \sum_k A_k}_{\text{correction to the ground state energy } H^{(0)}}$$

energy of zero-point motion

$$H^{(0)} + H^{(2)} = -J\, S(S+1) \frac{N}{2} \cdot z + JS \sum_k \omega_k \left(\alpha_k^+ \alpha_k + \tfrac{1}{2}\right) + \omega_k \left(\beta_k^+ \beta_k + \tfrac{1}{2}\right)$$

<u>Two</u> branches of spin waves : α_k and β_k.

$$\omega_k^2 = A_k^2 - B_k^2 = (2d)^2 - \left(2\sum_{i=1}^{d} \cos k_i a\right)^2 =$$

$$= 4d^2 \left\{ 1 - \left(\frac{1}{d} \sum_{i=1}^{d} \cos k_i a \right)^2 \right\}$$

<u>$d=3$</u> $\quad \omega_k = 6\sqrt{1 - \left(\frac{1}{3}(\cos k_x a + \cos k_y a + \cos k_z a)\right)^2}$

Minimum near $\vec{k} = \vec{0}$

$\cos k_i a \cong 1 - \frac{1}{2} k_i^2 a^2$

$$\omega_k \cong 6\sqrt{1 - \frac{1}{9}\left(3 - \frac{a^2}{2}(k_x^2 + k_y^2 + k_z^2)\right)^2} =$$

$$= 6\sqrt{1 - \left(1 - \frac{a^2}{6}\vec{k}^2\right)^2} = 6\sqrt{1 - 1 + \frac{2a^2}{6}\vec{k}^2} = \sqrt{12 a^2 \vec{k}^2}$$

$$= 12 a |\vec{k}|.$$

! * Spin-wave dispersion is linear in momentum ("relativistic"!)

Low-energy excitations: $\sum_k 12 a J S |\vec{k}| (\alpha_k^+ \alpha_k + \beta_k^+ \beta_k)$.

Ground state energy is obtained by taking expectation value $E_g = \langle H^{(0)} + H^{(2)} \rangle = -JS^2 \frac{N}{2} z - JS \frac{N}{2} z$

$$+ JS \sum_k \omega_k$$

because $\langle 0 | \alpha_k^+ \alpha_k | 0 \rangle = 0$ as $\alpha_k |0\rangle = 0$, $\beta_k |0\rangle = 0$.

$$E_g = -JS^2 \frac{N}{2} z \left\{ 1 + \frac{1}{S} - \frac{2}{SNz} \sum_k \omega_k \right\} =$$

$$= -JS^2 \frac{Nz}{2} \left\{ \underset{\substack{\uparrow \\ \text{classical}}}{1} + \frac{1}{S} \underbrace{\left[1 - \frac{2}{N} \sum_k \frac{\omega_k}{z} \right]}_{\text{quantum (first } \frac{1}{S} \text{ correction)}} \right\}$$

$$E_g = -\frac{1}{2} J S^2 N z \left\{ 1 + \frac{1}{S}\left[1 - \frac{2}{N}\sum_k \sqrt{1-\left(\frac{1}{d}\sum \cos k_i a\right)^2}\right]\right\}$$

Spin expectation value $\langle S_n \rangle = S - \langle a_n^+ a_n \rangle =$

$$= S - \frac{2}{N}\sum_{k_1 k_2} e^{i(\bar{k}_1 - \bar{k}_2)\bar{R}_n} \langle a^+_{k_1} a_{k_2} \rangle$$

$$\langle a^+_{k_1} a_{k_2} \rangle = \langle (ch\theta_{k_1} \alpha^+_{k_1} - sh\theta_{k_1} \beta_{-k_1})(ch\theta_{k_2}\alpha_{k_2} - sh\theta_{k_2}\beta^+_{-k_2}) \rangle$$

$$= ch\theta_{k_1} ch\theta_{k_2} \langle \alpha^+_{k_1}\alpha_{k_2}\rangle + sh\theta_{k_1} sh\theta_{k_2} \langle \beta_{-k_1}\beta^+_{-k_2}\rangle$$

$$- ch\theta_{k_1} sh\theta_{k_2} \underbrace{\langle \alpha^+_{k_1}\beta^+_{-k_2}\rangle}_{=0} - sh\theta_{k_1} ch\theta_{k_2} \underbrace{\langle \beta_{-k_1}\alpha_{k_2}\rangle}_{=0}$$

no anomalous terms in the ground state of α, β.

$\langle \alpha^+_{k_1}\alpha_{k_2}\rangle = 0$ because $\alpha_{k_2}|0\rangle = 0$ for any k_2.

But $\langle \beta_{-k_1}\beta^+_{-k_2}\rangle = \delta_{k_1,k_2} \langle \underbrace{\beta_{-k_1}\beta^+_{-k_1}}_{1+\beta^+_{-k_1}\beta_{-k_1}}\rangle = \delta_{k_1,k_2}$

$$\Rightarrow \langle S_n \rangle = S - \frac{2}{N}\sum_k sh^2\theta_k = S - \frac{2}{N}\sum_k \frac{1}{2}(ch 2\theta_k - 1)$$

$$= S - \frac{2}{N}\sum_k \frac{1}{2}\left(\frac{A_k}{\omega_k} - 1\right) = S\left\{1 - \frac{1}{2S}\cdot \underbrace{\frac{2}{N}\sum_k\left(\frac{1}{\sqrt{1-\gamma_k^2}} - 1\right)}_{\text{quantum correction } (1/S)}\right\}$$

classical result

$$\gamma_k \equiv \frac{1}{d}\sum_{i=1}^{d} \cos k_i a$$

Thus in $d=2$, for example, $\langle S_n \rangle = S\left\{1 - \frac{1}{2S}\cdot(1.393-1)\right\} =$

$= S\left(1 - \frac{0.393}{2S}\right)$ so that for $S = \frac{1}{2}$ the reduction is

$\langle S_n \rangle = \frac{1}{2}(0.607) = 0.303$

(about 40% reduction) instead of classical value $\frac{1}{2} = 0.5$